

Stopping the dots from flickering between letters

Written 2026-07-31. Design note - nothing here is built yet.

Mridul spotted this: when the cam turns from one letter to the next, it passes through every letter in between, so **the dots jump up and down repeatedly before settling**. For a blind reader resting a finger on the cell, that is garbage. It also looks wrong, and it hammers the linkages.

He proposed a **clutch** - drop the motor during rotation so the cam spins free, then re-engage.

This note answers: is the problem real, does that fix work, and what are the options.

1. Yes, the problem is real - here are the numbers

Counted from the actual encoding in `sim/braillix_params.json` (not estimated):

change	slices turned	individual dot movements
a -> b	4	7
h -> e	4	7
a -> t	18	37
e -> l	26	53
l -> o	28	56
a -> z	29	59

Typing "hello world":

359 dot movements actually performed
66 dot movements the job requires
5.4x wasted motion

Fifty-nine dot movements to change one character (a->z) is the worst case. Every one of those is a 1mm resin linkage being lifted against a spring.

Why it happens

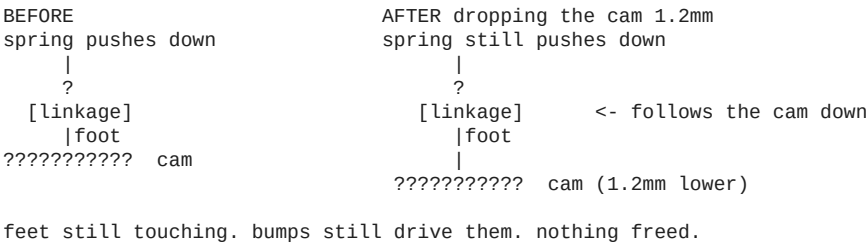
The cam is an **absolute position encoder** - each of the 64 angular slices *is* a letter. To reach a different letter you must physically rotate there, and the dots follow the tracks the whole way. This is inherent to a rotary absolute cam; it is not a bug in the profile.

Worth knowing: on a commercial multi-cell display an entire line refreshes while the reader's hand is elsewhere, so this matters less. On a **single-cell teaching device the finger is on the cell while it changes**, which is exactly why it matters here. Commercial displays did not solve this with a clutch - they use one latching actuator per dot.

2. The correction: dropping the motor does not free anything

This is the counter-intuitive part, and it changes the whole design.

Each return spring pushes its linkage **DOWN onto the cam**. So if the cam drops, the linkage simply follows it down. The foot never leaves the cam surface.



But this is **not** a dead end, because it still solves the reader's problem - it just solves it a different way than expected. That splits into two genuinely different mechanisms:

	What the reader feels	What it fixes	Complexity
Blanking - drop the cam ~1.2mm	Every dot goes below the reading surface. A clean blank cell, then the finished letter.	Reader experience [OK]	One moving assembly

	What the reader feels	What it fixes	Complexity
True declutch - add a hard down-stop, then drop the cam below it	Same, plus the feet genuinely leave the cam	Reader [OK] + wear [OK] + torque [OK]	Two mechanisms

Blanking needs no down-stop and no extra part beyond the drop itself, because the dots do not need to stop moving - they only need to stay below the surface where a finger cannot feel them.

The stated goal is the reader experience, so blanking is sufficient. Everything below is measured against that.

How far does it need to drop?

pin_lift is **0.8mm** (a raised dot stands 0.8mm proud). Drop the cam by more than that and even a fully-raised dot sits at or below the surface. **~1.2mm** gives 0.4mm of margin.

3. The space you have to fit it in

Read from the CAD, not from memory:

floor	z	0 .. 4	
electronics pocket	z	4 .. 18	14mm - the ULN2003 already lives here (36x46 pocket)
mid-plate ledge	z	18 .. 20	
mid-plate	z	20 .. 22	
MOTOR BODY	z	22 .. 41	19mm can (M2, measured)
base plate	z	41 .. 46	
cam disc	z	43 .. 45	+ 0.8mm bumps
linkage feet ride	z	45	
reading surface	z	57.2	

There is no spare volume. The electronics pocket is 14mm and mostly full. Any actuator has to earn its space from something that is already there.

4. The options

A - Drop the whole motor *(the original idea)*

Move the motor and cam together, 1.2mm.

- [OK] Conceptually simplest, cam/shaft alignment is untouched
- [X] You are lifting **~35g of motor** plus fighting the reaction of six springs (~3N)
- [X] Needs a hole through the mid-plate for the motor to descend into
- [X] **The motor wires must flex 1.2mm on every single character change** - that is a fatigue

failure waiting to happen at thousands of cycles

- [X] No actuator space left in the electronics pocket

B - Drop only the cam, on the shaft

Leave the motor bolted down; slide the **cam** along the shaft.

- [OK] **The double-D shaft is already a spline.** It transmits torque at any axial position - this is the single most useful fact in this note. The cam can slide up and down and still be driven.
- [OK] Moves ~4g instead of 35g. No wire flexing at all.
- [X] The hub is **dia 9 sitting inside a dia 10 bore** - there is **0.5mm of radial room**, nowhere near enough for a fork or yoke without enlarging the base-plate clearance hole
- [X] Still needs an actuator somewhere

C - Passive helical coupling - no second actuator * most elegant

Use **direction of rotation** as the control signal, with deliberate slop ("lost motion").

The cam is not rigidly keyed to the shaft. The coupling has a **helical slot** with ~20deg of play:

```

CLOCKWISE -----
pin climbs the helix -> cam pushed UP -> ENGAGED
pin reaches slot end -> cam now turns with the shaft

? ~20 degrees of lost motion
? (cam moves up/down but does NOT rotate)

```

ANTICLOCKWISE -----
pin descends the helix -> cam pulled DOWN 1.2mm -> BLANK
pin reaches slot end -> disc spins freely, dots stay sub-flush

To show a new letter:

- 1 Run **CCW** ~20deg -> cam drops, **dots go blank** (disc has not turned yet)
 - 2 Keep running **CCW** round to the target slice - the reader feels nothing
 - 3 Run **CW** ~20deg -> cam lifts, **the new letter appears**
- [OK] **No second actuator, no extra wiring, no extra current, no space cost**
 - [OK] Travel becomes one-way, which normally costs up to 63 slices instead of <=32 - but that is **free here**, because the dots are blanked during travel. It only costs time.
 - [X] Hardest thing in this note to design and to print. A helix in resin at dia 9 with 1mm walls.
 - [X] The 28BYJ-48 gearbox has significant backlash, which will interact with the lost-motion arc and must be characterised on the real motor

D - Lift the linkages instead [X] ruled out

Raising the followers off the cam raises **all six dots at once**. The reader would feel a full cell of raised dots during every transition - worse than the flicker. Rejected.

E - Sliding shutter over the reading surface

A thin plate slides across the dots during the change.

- [OK] Conceptually simple, and it is a pure reader fix
- [X] Must live in the **0.8mm finger-pad recess**. A plate plus its actuation in 0.8mm is not realistic at this scale.
- [X] Does nothing for wear or torque

F - Two cams, 8 states each - not a clutch, but nearly free ?

Already on the table as the likely fix for **R-07** (the cam pressure angle). Split six dots across two discs of 8 states each: 8 x 8 = 64 combinations.

Measured on the same "hello world" test:

one 64-state disc	359 dot movements
two 8-state discs	93 dot movements
	3.9x less chatter

- [OK] **No new mechanism at all** - it is a re-layout you may be forced into anyway
- [OK] Also fixes the pressure angle (9-12deg vs the current 71-79deg)
- [X] Costs a **second motor per cell**
- [X] Reduces the flicker ~4x, does not remove it

5. Actuator reality check

For options A, B and D you need something to do the moving:

	Size	Verdict
SG90 micro servo	23 x 12 x 29mm	[X] Will not fit beside the ULN2003 in a 36x46x14 pocket
Micro solenoid	~10 x 10 x 15mm	[!] Might fit. Costs a driver channel, continuous holding current, and heat
Second 28BYJ-48	dia 28 x 19mm	[X] No
Nitinol (muscle wire)	negligible	[!] Tiny, but slow and needs current control

This is the main argument for option C. It is the hardest to design and the only one that costs no space and no electronics.

6. Recommendation

1. Blanking is enough. The goal is the reader. A full declutch buys wear and torque savings that are not currently the problem, at roughly double the complexity.

2. Do not build any of this yet. Two reasons:

- **R-07 is unresolved.** If the cam ramp turns out to be unclimbable, the disc architecture changes and any clutch designed against the current 64-state disc is scrapped.
- **Option F may make it unnecessary.** If R-07 forces the two-cam split, chatter drops ~4x on its own. That may be good enough without any clutch at all.

3. The test that decides it costs two minutes. When the resin lands (~4 Aug), rest a linkage foot on a cam track and turn the disc by hand. That settles R-07, and R-07 settles which architecture the clutch would have to be designed for.

4. If it does get built, start from option C, and fall back to B with a micro solenoid if the helix proves unprintable.

Reproducing the numbers

Everything in section 1 comes from `sim/braillix_params.json`, which is itself generated from the CAD by `sim/extract_params.py`. Sweep each pair the short way round the disc and count the bits that flip between adjacent slices - a flipped bit *is* a dot moving:

```
def sweep(a, b, states=64):
    fwd, rev = (b - a) % states, (a - b) % states
    step, n = (1 if fwd <= rev else -1), min(fwd, rev)
    moves, cur = 0, a
    for _ in range(n):
        nxt = (cur + step) % states
        moves += bin(cur ^ nxt).count('1')
        cur = nxt
    return n, moves
```

The two anchors to check against: **a->z = 59 dot movements**, **"hello world" = 359**.